THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE CHAPTER TEST SET B

Chapter 2: **Exploring Mixtures and Their Separation** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. 20 g of salt is dissolved in 80 g of water. The mass by mass percentage of the solution is: 1

(a) 30 % **(b)** 25 % **(c)** 20 % **(d)** 10 %

2. 9 g of a solute is dissolved in water to make 1000 mL of solution. Its mass by volume percentage (% m/v) is: 1

(a) 0.89 % **(b)** 1.8 % **(c)** 0.45 % **(d)** 0.9 %

3. In a solution of sugar in water, the solvent is: 1

(a) sugar **(b)** water **(c)** both **(d)** neither

4. The process of forming pure crystals from a hot saturated solution on cooling is called: 1

(a) sublimation **(b)** chromatography **(c)** crystallization **(d)** distillation

5. Which technique is used to separate the different dyes present in black ink? 1

(a) distillation **(b)** sublimation
(c) decantation **(d)** paper chromatography

6. Dry ice is: 1

(a) solid carbon dioxide that sublimes **(b)** solid nitrogen that melts
(c) a compound of oxygen **(d)** frozen water

7. Which of these shows the Tyndall effect? 1

(a) sugar solution **(b)** clear vinegar
(c) salt solution **(d)** milk diluted with water

8. Metals can be mixed to form an alloy by: 1

(a) filtering **(b)** adding water
(c) stirring at room temperature **(d)** melting them together and cooling

9. The word chromatography comes from Greek words meaning: 1

(a) mixture and pure **(b)** heat and light
(c) colour and writing **(d)** water and oil

10. Crystallization differs from simple evaporation because it: 1

(a) needs no solvent
(b) always burns the solute
(c) gives pure crystals and removes soluble impurities
(d) removes only gases

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. What is a saturated solution? 2

12. Define crystallization and state its principle. 2

13. Why is a hot saturated solution cooled slowly during crystallization? 2

14. How can acetone be separated from a mixture of acetone and water? Give the principle. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Describe how copper sulfate crystals are prepared in the laboratory (four steps). 3

16. Describe the process of distillation with the help of the apparatus parts and name one use. 3

17. A student mixes acetone (b.p. 56 °C) and water (b.p. 100 °C). Explain how to separate them and why evaporation cannot be used. 3

OR

What is fractional distillation? Where is it used? 3

18. Explain paper chromatography with the steps to separate the dyes of a black sketch pen. 3

OR

Explain with one example why different components move up a chromatography paper at different speeds. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Distinguish between simple and fractional distillation. 5
(ii) Draw a labelled diagram of simple distillation (describe the parts).
(iii) Name two products of fractional distillation of petroleum and their uses.

OR

Case study: In Kannauj, the smell of the first rain on earth is captured as ‘mitti ka ittar’. 5
(i) Name the method used.
(ii) Explain its principle.
(iii) Name one other use of the method.
(iv) Why can the product not be obtained by filtration?

20. (i) What is paper chromatography? 5
(ii) Describe the experiment to separate the pigments in black ink.
(iii) What would you observe and why?
(iv) Why must the solvent level be below the spot?

OR

(i) Distinguish between a solution, a colloid and a suspension with examples. 5
(ii) State the Tyndall effect.
(iii) How would you show that milk is a colloid?

END OF QUESTION PAPER — Total: 40 Marks